

VoxShield Edge

Real-Time, Privacy-Preserving Voice-Clone Defense for Indian Telephony

A bank fraud analyst's HP Snapdragon laptop detects AI-cloned voices in real time, fully on-device on the Hexagon NPU — no call audio ever leaves the machine.

5.9% EER · deployed fusion (held-out In-the-Wild)	6.8% EER · unseen-generator deepfakes (OOD, MLAAD)	0.98 ROC-AUC on OOD deepfakes	45–80 TOPS Hexagon NPU (Snapdragon X / X2)
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Individual submission · Devansh Goenka · GitHub: DG10911/voxshield

■ Problem

AI voice cloning is cheap and fast. In India, phone-based fraud — fake bank officials, “digital arrest” scams, relative-in-distress calls — increasingly uses synthetic voices in Indian languages over ordinary 8 kHz phone lines. A verdict after the call ends is forensics, not protection — and sensitive customer call audio should not be shipped to a foreign cloud.

■ Solution — VoxShield Edge

VoxShield Edge is an **on-device, Indic-first, telephony-hardened** voice-deepfake detector for banking. Given a live call it returns a calibrated **LOW / MEDIUM / HIGH** risk verdict with explainable reason codes — decision support for a fraud analyst, **never automatic blocking**. It runs entirely on a Snapdragon-powered HP laptop, so raw call audio never leaves the device.

■ Technical implementation

A streaming pipeline: 8 kHz G.711 telephony audio → Silero VAD → 3 s windows at a 1 s hop → an 89-dimensional acoustic feature bank (computed at a 16 kHz analysis rate) → **five complementary detectors** (Acoustic DSP · wav2vec2 · XLS-R 300M · DistilHuBERT · LFCC+CQCC) → a **learned meta-stacker** with Indic / telephony routing → Platt-calibrated risk + per-signal reason codes (spectral, phase, prosody, breath).

■ Snapdragon optimisation & on-device deployment

VoxShield Edge is built to run on the Snapdragon Hexagon NPU (45 TOPS on Snapdragon X Elite/Plus, 80 TOPS on X2 Elite). The on-device path: **PyTorch** → **ONNX (INT8-quantised)** → **ONNX Runtime with the QNN Execution Provider (onnxruntime-qnn)** → **Hexagon NPU**, packaged as a Windows/ARM64 app. Models are compiled and profiled on real Snapdragon devices via **Qualcomm AI Hub (qai-hub)**, and the design leverages the AI Hub speech ecosystem — **WavLM-Base-Plus** (a self-supervised speech backbone) and **Whisper** for on-device transcription. **Honest status:** the detector runs today on CPU/GPU; the NPU port and INT8 quantisation via AI Hub are the optimisation in progress — no on-NPU latency is claimed until measured.

■ Evidence (measured, honestly scoped)

Model / setting	Evaluation	Metric	Result
Deployed fusion	held-out In-the-Wild	EER	5.9%
Unseen-generator deepfakes	MLAAD (out-of-distribution)	EER / AUC	6.8% / 0.98
Detector scoreboard	same eval set	EER single/avg/fusion	16.0 / 19.1 / 5.9
Indic recall	MMS-TTS fakes (after retrain)	recall	42% → 82%

EER (Equal Error Rate) is not accuracy; in-corpus and out-of-distribution figures are reported separately and never conflated. The 6.8% EER on unseen generators is the honest cross-generator generalisation result.

■ Innovation, accessibility & impact

- **Innovation:** the integration under one hard constraint set — Indic language + narrowband telephony + on-device NPU + explainable + private — not a single new algorithm.
- **Accessibility:** offline, low-power on the NPU, works in poorly-connected branches; no cloud cost per call.
- **Privacy:** no raw-audio egress — architected to support institutional data-governance requirements.
- **Impact:** a private, on-device fraud-screening layer any bank can run on a Snapdragon laptop.

GitHub: DG10911/voxshield · Pitch deck (PDF + PPTX) submitted separately · Contact: devanshgoenka03@gmail.com